

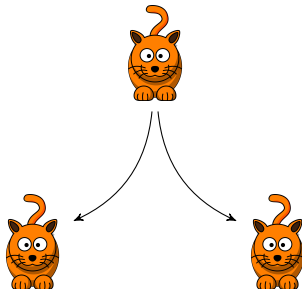
Non-regular Languages

Alessandro Chiesa



School of Computer and Communication Sciences

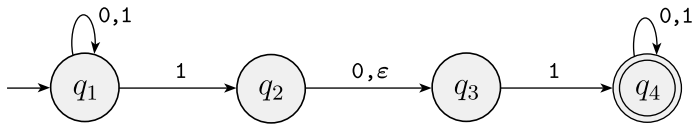
Lecture 3



Should I stay or should I go (left or right)

RECALL: NFAs, Subset Construction

Nondeterminism vs Determinism



State diagram of a **Nondeterministic Finite Automaton (NFA)**

Differences to DFAs:

- 1 Ability to transition to **more than one** state on a given symbol
Ex: q_1 has two outgoing transitions for symbol 1
- 2 A state may have **no transition** on a particular symbol
Ex: q_3 has no outgoing transitions for symbol 0
- 3 Ability to take a step **without** reading any input symbol (ϵ -transitions)
Ex: q_2 can transition to q_3 without reading a symbol

Formal definitions

A **nondeterministic finite automaton (NFA)** M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where

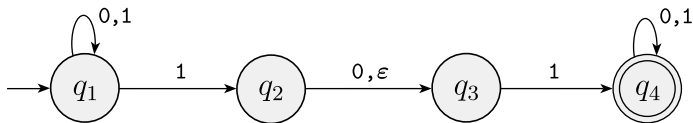
- ▶ Q is a finite set called the **states**,
- ▶ Σ is a finite set called the **alphabet**,
- ▶ $\delta : Q \times (\Sigma \cup \{\varepsilon\}) \rightarrow 2^Q$ is the **transition function**,
- ▶ $q_0 \in Q$ is the **start state**, and
- ▶ $F \subseteq Q$ is the **set of accept states**. (allow $F = \emptyset$)

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- ▶ Here 2^Q denotes the power set of Q . Ex:
 $Q = \{q_1, q_2\}, 2^Q = \{\emptyset, \{q_1\}, \{q_2\}, \{q_1, q_2\}\}$
 - ▶ An input is accepted if there **exists at least one path** that ends at an accepting state
 - ▶ An input is rejected if **no computation path** ends at an accepting state

Example



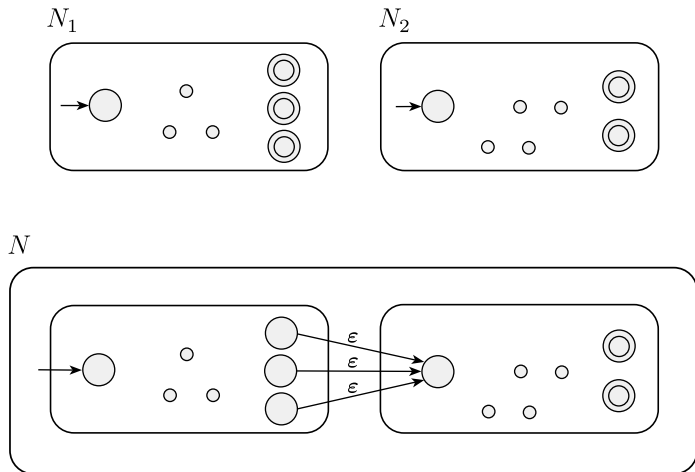
The formal description of N_1 is $(Q, \Sigma, \delta, q_1, F)$, where

1. $Q = \{q_1, q_2, q_3, q_4\}$,
2. $\Sigma = \{0,1\}$,
3. δ is given as

	0	1	ϵ
q_1	$\{q_1\}$	$\{q_1, q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\emptyset$	$\{q_3\}$
q_3	$\emptyset$	$\{q_4\}$	$\emptyset$
q_4	$\{q_4\}$	$\{q_4\}$	$\emptyset,$

4. q_1 is the start state, and
5. $F = \{q_4\}$.

Concatenation





Theorem. Every nondeterministic finite automaton has an equivalent deterministic finite automaton.

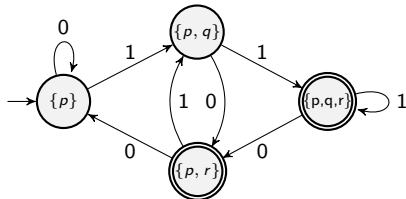
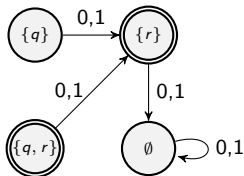
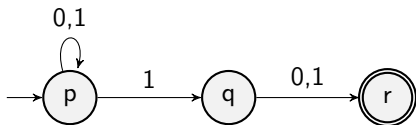
Corollary. A language is regular if and only if some nondeterministic finite automaton recognizes it.

Example

$$\tilde{\delta}: 2^Q \times \Sigma \rightarrow 2^Q$$

$$\tilde{\delta}(A, a) = \bigcup_{q \in A} \delta(q, a)$$

$\tilde{\delta}$	0	1
$\emptyset$	$\emptyset$	$\emptyset$
$\{p\}$	$\{p\}$	$\{p, q\}$
$\{q\}$	$\{r\}$	$\{r\}$
$\{r\}$	$\emptyset$	$\emptyset$
$\{p, q\}$	$\{p, r\}$	$\{p, q, r\}$
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Is every Language Regular?

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- ▶ Pattern matching

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- ▶ Skipping prefixes, suffixes, ... (e.g. 2nd last bit is 1)

More generally – “regular” expressions

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More generally – “regular” expressions

- ▶ $L = 0^*110^* - \{0^n110^m \mid n, m \in \mathbb{N}\}$

- ▶ Closure under complements, unions, intersections, ...

Test your intuition

Are these languages regular?

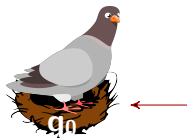
$$B = \{0^n 1^n \mid n \geq 0\} = \{\varepsilon, 01, 0011, 000111, \dots\}$$

$$C = \{w \mid w \text{ has an equal number of 0s and 1s}\}$$

$$D = \{w \mid w \text{ has an equal number of occurrences of 01 and 10 as substrings}\}$$

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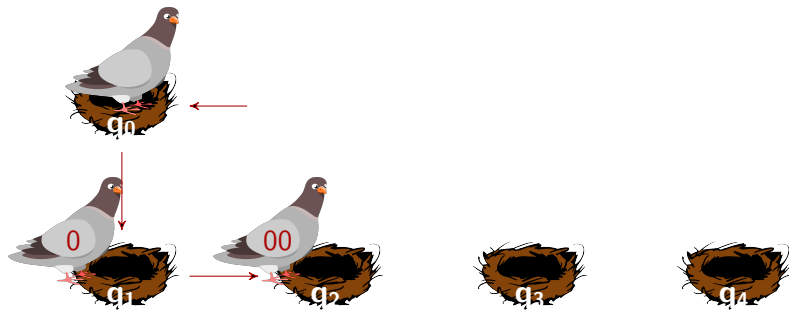
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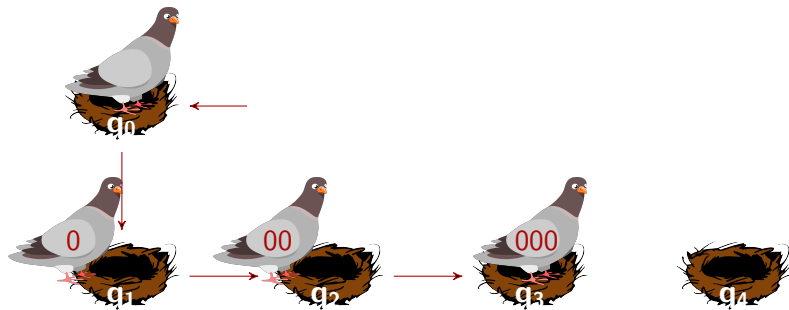
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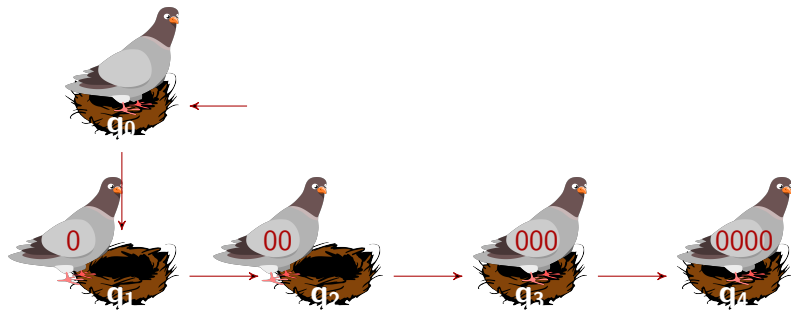
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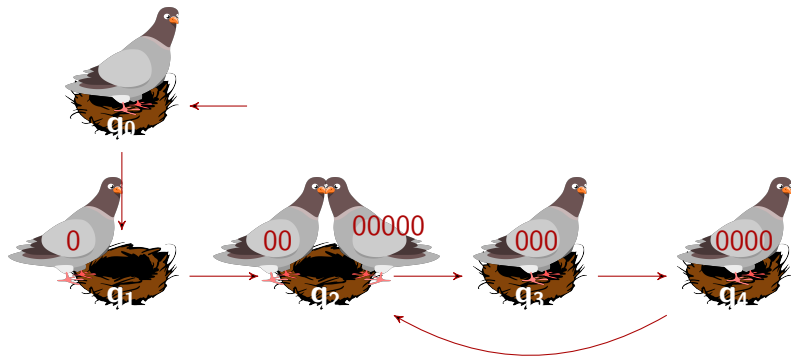
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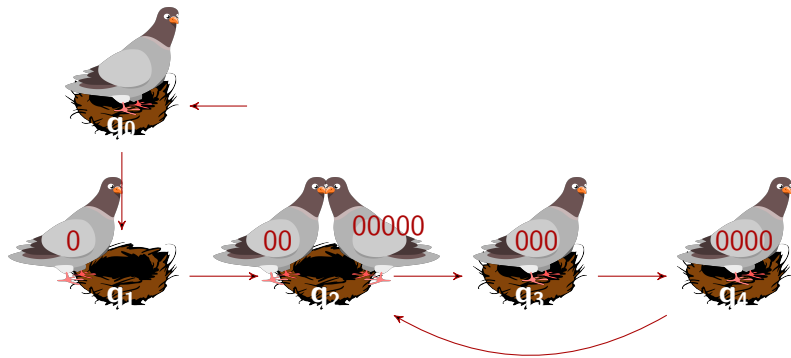
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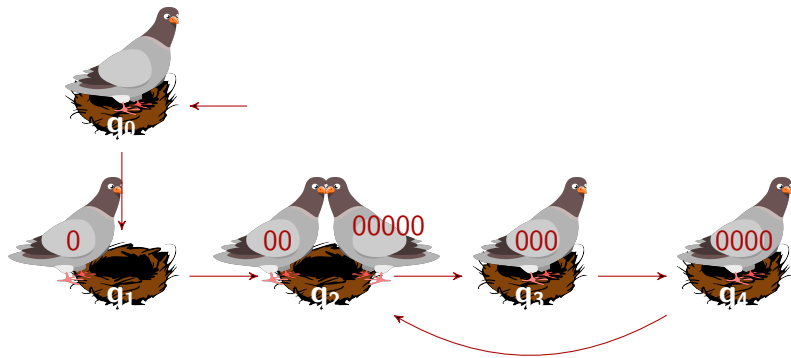
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- ▶ Thus, for two strings $0^i, 0^j, i \neq j$ we must end up in the same state

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- ▶ Thus, for two strings $0^i, 0^j, i \neq j$ we must end up in the same state
- ▶ Since $0^i 1^i$ is accepted $0^j 1^i$ is also accepted – contradiction!

Thus B is NOT regular!

Pumping Lemma and its Proof Sketch

If A is a regular language, then there is a number p (the pumping length) such that, for every string s in A of length at least p , there exists a division of s into three pieces, $s = xyz$ s.t.

1 for each $i \geq 0$, $xy^iz \in A$

2 $|y| \geq 1$, and

3 $|xy| \leq p$.

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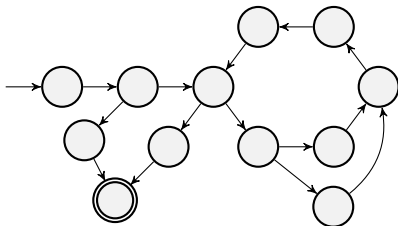
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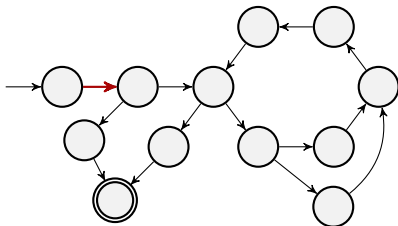
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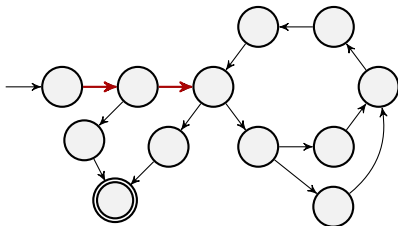
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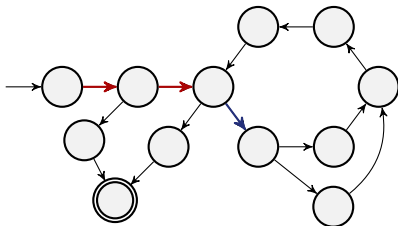
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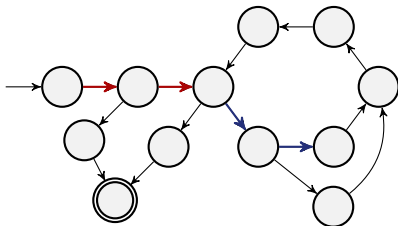
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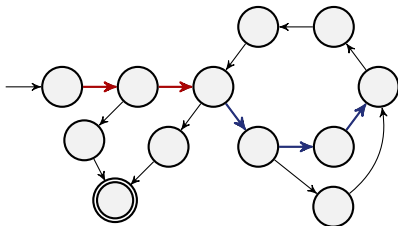
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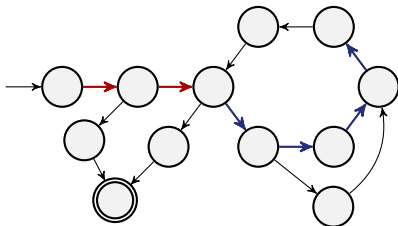
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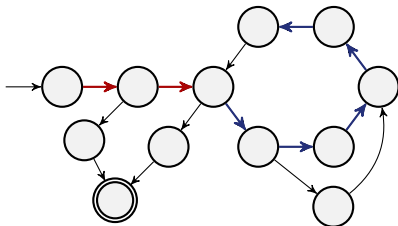
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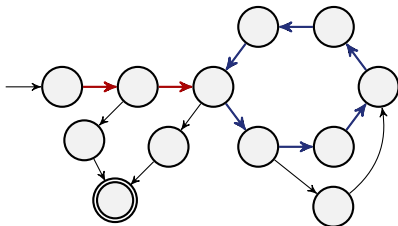
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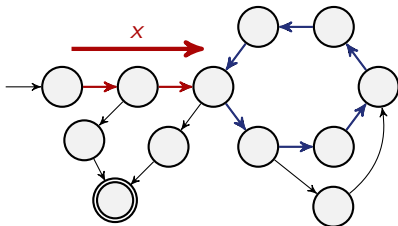


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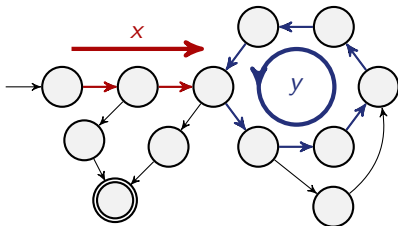
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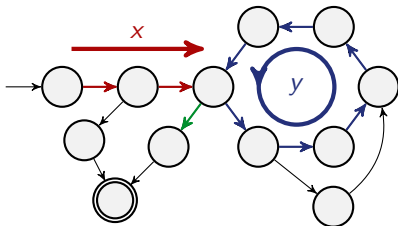
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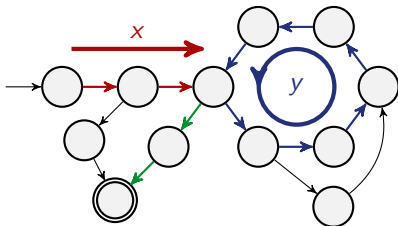
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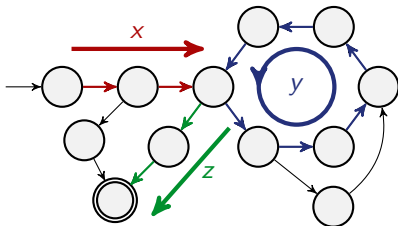


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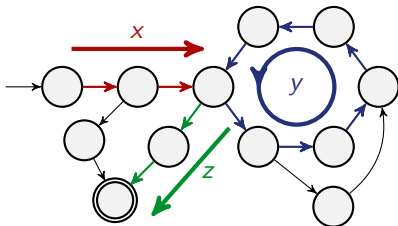
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(What if A is finite?)

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More formal proof

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- ▶ Let $p = |Q|$ be the number of states in a DFA $M = (Q, \Sigma, \delta, q_0, F)$ accepting A
- ▶ Let s be an accepted string of length $|s| \geq p$
- ▶ Let q be the first state that is repeated when starting at q_0 and feeding s and let x, y, z be the strings s.t. $|xy|$ is minimized and

$$\delta(q_0, x) = q \quad \text{and} \quad \delta(q_0, xy) = q \quad \text{and} \quad \delta(q, z) = r \in F$$

- ▶ Then $\delta(q_0, xy^iz) = r$ for all $i \geq 0$
- ▶ Note that $|y| \geq 1$ as there are two distinct occurrences of q
- ▶ $|xy| \leq p$ by pigeonhole principle and that $|xy|$ equals the number of steps until a state is visited twice.

Applications

$$F = \{ww \mid w \in \{0,1\}^*\}$$

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Pumping lemma tells us there is such a decomposition – we can't choose it! Your reasoning should work for any decomposition

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- ▶ However, $xy^2 z = 0^{p+k+1}1^p$, $xy^3 z = 0^{p+2k+1}1^p, \dots \in E$

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- ▶ But $xz = 0^{p+1-k}1^p$ and $p+1-k \leq p$. Thus, $xz \notin E$.
Contradiction!

$$C = \{w \mid w \text{ has equal number of 0s and 1s}\}$$

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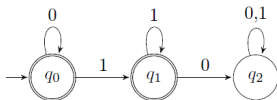
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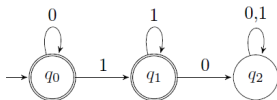
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Proof: (by contradiction)

- ▶ Assume C is regular
- ▶ Let $0^* 1^* = \{0^m 1^n \mid m, n \geq 0\}$
- ▶ $0^* 1^*$ is regular
- ▶ Claim: $B = C \cap (0^* 1^*)$



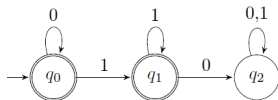
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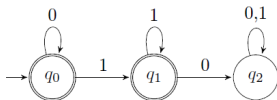
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- ▶ Intersection of two regular languages is regular
- ▶ $\Rightarrow B$ is regular – a **contradiction!**



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 - ▶ **Contradiction!**

Task: Prove that L is not regular ..

Let's try to use the Pumping Lemma



Assume L is regular



p : the pumping length

Fun part is
guessing which
string to pick!

Your reasoning should
work for any
decomposition

Pick s in L with $|s| \geq p$

$s = xyz$ s.t.

$|xy| \leq p$ and $|y| \geq 1$ and $xy^i z \in L \quad \forall i \geq 0$

$xz, xyyz, xy^2yz, xy^3yz, \dots$

If one of them NOT in L – DONE!

If all of them in L – Try again!

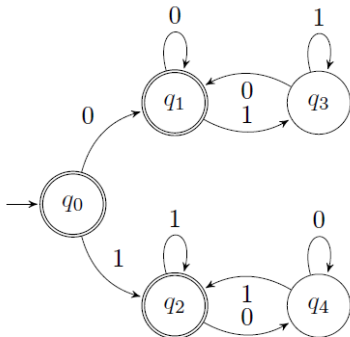
Test your intuition...

$$D = \{w \mid w \text{ has equal nr of occurrences of } 01 \text{ and } 10 \text{ as substrings}\}$$

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D is regular!



Beyond Regular Languages

Turing Machines . . .